Hardware-Aware Training and Precision-Retention Frontiers for Low-Precision Analog Vision

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Abstract

We present a behavioral simulation framework for analog compute-in-memory inference, focusing on precision-retention trade-offs under realistic physical deployment constraints. Under a tested AIHWKit IdealDevice noise baseline, standard fixed-instance hardware-aware training achieves 87.28% fresh-instance accuracy at 8-bit precision but collapses to 14.64% at 4-bit. We show that Ensemble Hardware-Aware Training (HAT), which resamples device-to-device mismatch masks during training, rescues the 4-bit pure-quantization regime, recovering fresh-instance accuracy to 86.16%. We then evaluate a realistic phase-change memory (PCM) simulation regime. Under the tested PCM UnitCell, models converge across 4-bit, 6-bit, and 8-bit precisions, but retention behavior is precision dependent. The results reveal a precision-retention deployment frontier: 8-bit PCM is drift-flat over 24 hours, while 4-bit PCM reaches the most aggressive tested compression but incurs a measurable 4.01 pp time-dependent drift. In this tested sweep, 6-bit PCM is the best observed Pareto midpoint, maintaining 8-bit-like drift stability while achieving comparable fresh accuracy. These findings separate the algorithmic challenge of cross-instance robustness from the physical constraints of analog retention.

Keywords: analog compute-in-memory; hardware-aware training; phase-change memory; low-precision inference; precision-retention tradeoff; vision transformers; mixed-signal simulation.

1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications directly within memory arrays, eliminating the data-movement bottleneck that limits conventional digital systems[1–3]. Across emerging analog memories, including resistive memories, phase-change memory (PCM), and organic optoelectronic devices, deployment accuracy is shaped by finite conductance precision, device-to-device variability, cycle-to-cycle noise, and retention drift[4–8]. These effects are often characterized at the device level, but their system-level consequences for modern vision backbones remain difficult to rank before fabrication.

However, translating device-level models into deployment-facing model accuracy faces a critical methodological gap. Mature CIM simulators expose increasingly detailed analog effects[2;4;5;9], yet the algorithmic and physical failure modes are often entangled: a low-precision model can fail because hardware-aware training overfits one fixed mismatch realization, because quantization is too coarse, or because a realistic memory substrate drifts after programming. Device-specific studies, including organic and optoelectronic demonstrations, further highlight this gap because variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are frequently absent from network-level analyses or treated only qualitatively[10–12]. A deployment workflow therefore needs to separate algorithmic robustness from physical precision-retention limits.

Here we present a behavioral simulation framework that evaluates analog neural network deployment under both idealized pure-quantization and realistic device-physics regimes. We apply a mixed-signal deployment model to the Tiny-ViT-5M backbone to systematically disentangle algorithmic generalization failures from physical retention limits. Three concrete contributions

follow. First, we expose a fresh-instance transfer failure in standard fixed-instance hardware-aware training at extreme low precision (4-bit), where accuracy collapses to near-chance. We introduce Ensemble Hardware-Aware Training (HAT), which resamples device-to-device mismatch masks each epoch, rescuing the 4-bit pure-quantization regime and restoring fresh-instance accuracy to 86.16±0.19%. Second, we translate this algorithmic rescue into a realistic phase-change memory (PCM) simulation regime and observe stable convergence under the tested PCM UnitCell setting where pure quantization fails. Third, we map a 4/6/8-bit precision-retention deployment frontier for PCM. In this tested sweep, 4-bit is trainable but drift-limited, 8-bit is drift-flat, and 6-bit provides the best observed Pareto midpoint between model compression and long-term inference reliability.

The framework is intentionally behavioral rather than circuit-accurate, focusing on macroscopic precision and drift phenomena rather than array-level parasitics (IR drop, sneak paths) which remain crucial future considerations. Additional supporting analyses, including front-end compensation and zero-shot transfer to alternative organic optoelectronic profiles, are provided in the Supplementary Information. Its purpose is to provide a reproducible workflow for comparing mitigation strategies and establishing physical deployment frontiers.

2 Related Work

2.1 Hardware-Aware Training and Robustness

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization[13]. Although related in spirit to quantization-aware training[14] and to domain randomization in sim-to-real transfer[15], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for a given hardware instance. It is therefore different from independent thermal or sampling noise. Ensemble HAT resamples the static D2D mask at each training epoch, improving zero-shot transfer to fresh hardware instances. The connection to domain randomization is deliberate, but the analogy is incomplete because hardware-induced perturbations decompose into D2D mismatch—spatially structured and fixed for a given fabricated instance—and cycle-to-cycle (C2C) noise—stochastic per-read with zero mean.

2.2 Organic Neuromorphic and Optoelectronic Devices

Organic synaptic and memristive devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware[6;8;16;17]. Recent demonstrations include multilevel memory[7;11], long retention tails[10], and integrated optoelectronic arrays[18–21]. Array-level studies are beginning to treat active-matrix addressing, spatial optical response, and crosstalk as system constraints rather than purely materials observations[22–24], while temperature-resilient organic synapses and high-temperature synaptic phototransistors suggest that thermal stability is also becoming deployment-relevant[25;26].

2.3 CIM Simulation Frameworks and Hybrid Mapping

System-level CIM simulators such as DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy[2]. MemTorch showed how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation[9], AIHWKIT exposed analog HAT and inference simulation in a practical mixed-signal software stack[4], and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models[5]. These mature simulators were developed primarily around inorganic resistive memories, and they do not directly expose all profile-level effects of interest, such as inverse-gamma optoelectronic photoresponse or

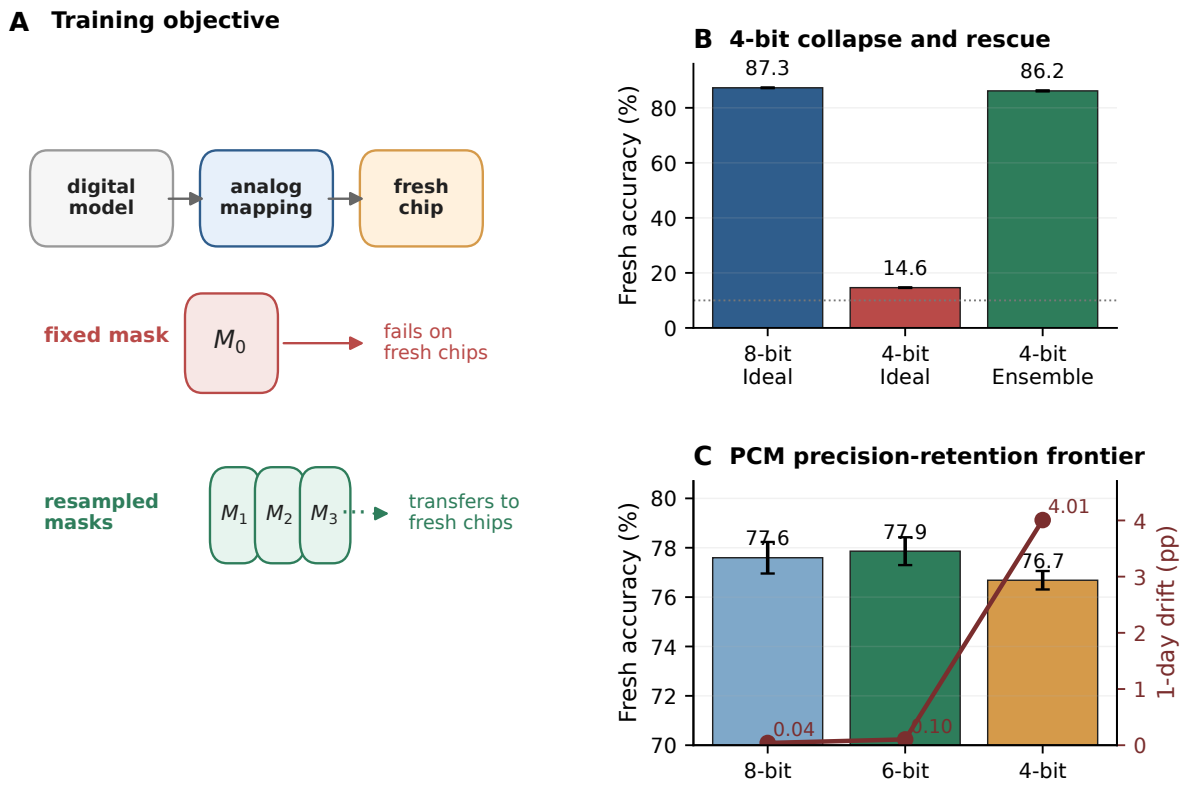

Figure 1: Paper-1 experimental spine. (A) Standard fixed-mask HAT optimizes one device instance, whereas Ensemble HAT trains over resampled D2D masks. (B) In the idealized AIHWKit baseline, 4-bit fresh-instance accuracy collapses while Ensemble HAT recovers it. (C) Under PCM UnitCell simulation, 4/6/8-bit models all train, but 4-bit has larger one-day drift and 6-bit is the best tested precision-retention midpoint.

organic-specific double-exponential retention, as first-class device primitives. Our behavioral profile-driven approach therefore serves as a profile-level extension and deployment-risk ranking layer rather than as a replacement for established CIM toolkits.

3 Results

3.1 The Failure of Pure Quantization and Algorithmic Rescue

We first measure the behavior of the tested AIHWKit IdealDevice quantization/noise baseline. At 8-bit precision the baseline remains stable, achieving 87.28 ± 0.13% fresh-instance accuracy; at 4-bit precision it collapses to 14.64 ± 0.11%. This indicates that standard fixed-instance hardware-aware training fails to generalize under fresh hardware instance shifts at extreme low precision in this tested regime.

To address this, we apply Ensemble Hardware-Aware Training (HAT). By resampling the device-to-device (D2D) variation patterns during the forward pass, Ensemble HAT rescues the tested 4-bit pure-quantization regime where the AIHWKit IdealDevice baseline collapses. The model recovers to 86.16 ± 0.19% fresh-instance accuracy, effectively closing the gap with the 8-bit baseline and demonstrating robust cross-instance transfer.

3.2 Realistic PCM Training and the Precision-Retention Frontier

While Ensemble HAT addresses the algorithmic generalization problem in a pure-quantization context, realistic analog deployment must account for complex device physics, particularly

conductance drift. We evaluate our approach using the `PCMPresetUnitCell` model in AIHWKit, which incorporates non-linear update physics, finite dynamic range, and time-dependent drift.

Under the tested PCM UnitCell simulation regime, training is viable across 4-bit, 6-bit, and 8-bit precisions. However, the results reveal a precision-retention deployment frontier rather than a single best compression point. Figure 2 converts the result into a deployment decision map: distribution-level HAT fixes the fresh-instance failure, while PCM precision selects the retention budget.

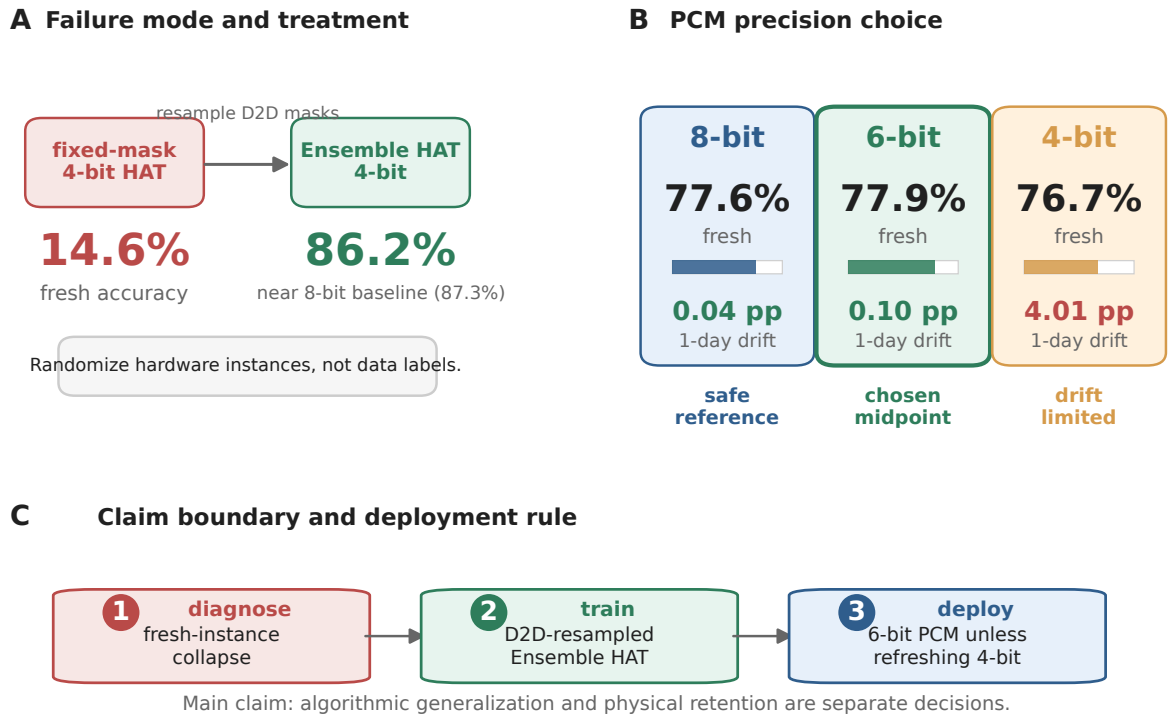

Figure 2: Results-to-deployment decision map. (A) Fixed-mask 4-bit HAT collapses under fresh hardware instances, while D2D-resampled Ensemble HAT restores 4-bit fresh accuracy near the 8-bit idealized baseline. (B) In PCM UnitCell simulation, 8-bit and 6-bit are drift-flat over one day, whereas 4-bit is trainable but drift-limited. (C) The resulting deployment rule is sequential: diagnose fresh-instance collapse, train with a mismatch-distributed objective, then choose the PCM precision from the retention budget.

3.3 Deployment Guidance

The precision ladder separates the algorithmic challenge of cross-instance robustness from the device-level precision-retention tradeoff. 8-bit PCM is the drift-flat reference; 4-bit PCM is trainable but loses ∼4.01 pp over one day; 6-bit is the best observed midpoint in the tested PCM UnitCell setting, with $77.86 \pm 0.56\%$ fresh accuracy and only 0.10 pp one-day drift.

3.4 Future Directions

Extending the same hardware-aware training principles to memory-bound inference components such as analog KV-cache is an important future direction.

4 Discussion

4.1 Diagnosis: hardware-instance overfitting as the leading tested failure mode

Once hardware-aware training (HAT) and scale recovery are available, standard cycle-to-cycle (C2C) and device-to-device (D2D) variability do not dominate the error budget. Three constraints dominate instead.

First, readout precision remains a hard gate: below 6 bits, transformer inference degrades substantially. Second, the leading tested training failure is hardware-instance overfitting. Standard HAT fits one fixed D2D realization and collapses to 10.00% on fresh instances, whereas the mismatch-distributed objective in Eq. 2 raises fresh-instance accuracy to $86.16 \pm 0.19\%$ across three seeds. Third, once this generalization problem is controlled, realistic PCM simulation exposes a separate precision-retention frontier rather than a single best compression point.

This ordering matters for design. The paper's main claim is not that one training recipe solves all device physics; it is that cross-instance robustness and physical retention should be solved in sequence. A residual gap to the 98.06% digital baseline remains, and capacity-aware Ensemble HAT schedules are left as an open optimization problem.

4.2 Treatment: Ensemble HAT mitigation across scenarios

Ensemble HAT should be used when the observed failure mode is fresh-instance transfer under a low-precision analog-noise model. In that regime, epoch-level D2D resampling converts a 4-bit collapse into near-8-bit fresh-instance accuracy. It should not be read as a complete device-physics solution. Under the tested PCM UnitCell regime, 4-bit remains trainable but is drift-limited, whereas 6-bit and 8-bit are effectively drift-flat over the tested 24-hour window.

4.3 Mechanism: theory and empirical landscape

Supplementary Note S-Theory gives the formal view: per-epoch resampling is, to second order, a weighted gradient-L_2 regularizer and a stochastic sharpness penalty along the device-mismatch direction. The PAC-Bayes argument explains why fresh-instance accuracy should track the training-distribution average when the training distribution spans the deployment mismatch.

The empirical diagnostics agree with this view. Loss interpolation from the training D2D mask toward fresh masks shows a $\sim$76 pp Standard-to-Ensemble gap, and the extrapolated $3\times$ mismatch test preserves a 17.5 pp advantage for Ensemble HAT. Hessian spectra add an important caution: Standard HAT can appear flatter at its own training mask, but that flatness is mask-specific. The relevant robustness metric is therefore D2D-directional sensitivity, not global curvature.

4.4 Implications: design rules for deployment

Figure 2 gives the deployment order used by this work: diagnose fresh-instance collapse, train with a mismatch-distributed objective, then select the physical precision from the retention budget. In the tested PCM UnitCell setting, 4-bit is attractive for compression but drift-limited, while 6-bit is the safest observed midpoint. Detailed ADC/D2D envelope and front-end compensation checks are retained in the Supplementary Information rather than elevated to additional main-text claims.

4.5 Comparison to established analog HAT primitives

AIHWKit[4] per-batch noise injection is already strong at 8-bit ($87.28 \pm 0.13\%$ fresh-instance), but it collapses at 4-bit ($14.64 \pm 0.11\%$). Ensemble HAT maintains $86.16 \pm 0.19\%$ at 4-bit. Its

advantage is therefore specific to the operating point where per-batch noise injection alone does not overcome mismatch variability. The PCM results then add a second constraint: all tested precisions train, but only 6-bit and 8-bit remain drift-flat over the tested window.

4.6 Limitations and outlook

Scope. The conclusions are bounded by the present behavioral simulation stack. Device profiles are calibrated from literature values or a single fabrication batch, so the numbers do not claim cross-batch, temperature, aging, or circuit-layout coverage. We model quantization, D2D/C2C variability, ADC precision, write nonlinearity, and retention, but defer parasitic IR drop, sneak paths, and pulse-faithful programming to follow-on hardware-calibrated studies.

Energy and scale. The energy estimates are first-order analytical projections from literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ; Section 5.5). They rank deployment risks rather than predict silicon performance. The current ablation matrix also prioritizes non-ideality coverage over dataset breadth; cross-architecture TinyImageNet sweeps are treated as validation, not as main-text claims.

Outlook. The next tests are straightforward: replace UnitCell presets with measured-array statistics, verify whether the 6-bit midpoint persists, and evaluate memory-bound inference components such as analog KV-cache with explicit train/eval noise separation. These extensions do not alter the present conclusion: algorithmic cross-instance resampling and physical precision-retention selection are separate deployment decisions.

5 Methodology

5.1 Hybrid Analog/Digital Deployment

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Concretely, the analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers; the digital partition contains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token, following established practice in heterogeneous CIM accelerators[27–31].

Under this mapping, approximately 88% of parameters execute on analog arrays while roughly 60% of estimated energy remains digital, largely because attention operations stay in the digital domain—a pattern consistent with recent heterogeneous CIM accelerators for Vision Transformers[28–31].

5.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window $[G_{\min}, G_{\max}]$, and quantized to n_{states} levels via a straight-through estimator (STE) quantizer $Q_n(\cdot)$. Differential readout recovers $G_{\text{eff}} = \tilde{G}^+ - \tilde{G}^-$, and the downstream digital weight is

$$\hat{W}_{ij} = s_\ell \, G_{\text{eff},ij}, \qquad s_\ell = \begin{cases} \dfrac{w_{\max}}{G_{\max} - G_{\min}}, & \text{standard calibration} \\[2ex] \dfrac{w_{\max}}{\max_{i,j} |G_{\text{eff},ij}| + \epsilon}, & \text{retention-time recalibration} \end{cases} \tag{1}$$

with layer index ℓ omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for the V8 retention-drift experiments. The recovered weight step is $\Delta_{\hat{W}} = s_\ell(G_{\max} - G_{\min})/(n_{\text{states}} - 1)$, so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise[32;33].

Hardware-aware training (HAT) injects forward perturbations while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field M_0 during training.

Ensemble HAT as distribution matching. Ensemble HAT treats the mismatch map as a random variable resampled each epoch:

$$\theta^\star_{\text{ens}} = \arg\min_\theta \; \mathbb{E}_{M\sim\mathcal{D}_{\text{D2D}}}\Big[\frac{1}{N}\sum_{n=1}^{N}\ell\big(f(x_n;\theta,M),y_n\big)\Big], \tag{2}$$

where the expectation is Monte-Carlo estimated with one fresh draw $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ held constant across all mini-batches in epoch e. Expanding Eq. 2 (Supplementary Note S-Theory) reveals an implicit Fisher-weighted gradient-squared regularizer that pushes the optimizer toward flat regions in the M-direction, explaining the fresh-instance transferability without extra hyperparameters. Epoch-level (not per-batch) resampling is load-bearing, giving the optimizer enough consistent signal to fit each instance before resampling[15]. Fresh-instance transfer is then evaluated on unseen i.i.d. D2D draws held constant for each evaluation pass. For a single checkpoint, the reported fresh-instance spread is the standard deviation across ten fresh-instance means (each averaged over five forward passes); for the R10A reproducibility headline, the uncertainty is the sample standard deviation across three seed-level fresh-instance means.

5.3 Training Protocol

All Tiny-ViT experiments use AdamW ($lr = 5\times10^{-4}$, weight decay 0.05), 100 epochs, batch size 64, cosine annealing ($T_{\max} = 100$). Images are resized to 224×224 with standard augmentation. The default analog resolution is $n_{\text{states}} = 16$ levels. V1 is digital; V2 is hybrid without noise; V3 is hybrid with fixed D2D ($\sigma_{\text{D2D}} = 10\%$); V4–V7 use full HAT with epoch-resampled D2D and per-forward C2C ($\sigma_{\text{C2C}} = 5\%$ or 10%). V6 adds the photoresponse front end; V7 is a legacy retention-aware model excluded from canonical claims; V8 is the corrected retention-aware follow-up used for retention checks.

Nonlinear-write and retention surrogates are anchored to measured DNTT transients[34]. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent factor proportional to the present conductance magnitude raised to $NL - 1$[1]; ideal write is recovered when $NL_{\text{LTP}} = 1$ (long-term potentiation) and $|NL_{\text{LTD}}| = 1$ (long-term depression). The optional optoelectronic front-end model (V6) applies inverse-gamma preprocessing, photocurrent shot-noise injection, and per-sample renormalization; the explicit response equations are kept in the Supplementary Information because V6 is not part of the main canonical claim.

5.4 Profile-Driven Substitution Interface

The profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle. Deployment holds the digital backbone and training recipe fixed while substituting the profile, enabling literature priors today and measured-device calibration later without code changes.

[1]The absence of an explicit NL prefactor is intentional: this is a behavioral proxy for position-dependent update difficulty, not the strict derivative of a power-law $f(G) = G^{NL}$.

Table 1: Notation used for the Tiny-ViT experiment family in the main text.

ID	Meaning
V1	FP32 digital baseline
V2	Quantized hybrid control without analog noise
V3	Standard training with a fixed D2D realization; noisy deployment baseline
V4	Uniform-noise hardware-aware training result used as the canonical deployment model
V5	Pessimistic robustness stress test with larger uniform noise
V6	V4 plus physical front-end compensation
V7	Legacy retention-aware model, excluded from the canonical claims
V8	Corrected retention-aware follow-up used only for retention checks

5.5 Sensitivity and Energy Metrics

For the joint D2D–ADC sweep, first-order factor importance is quantified with Sobol indices estimated from the 7×9 grid of Monte Carlo means. Energy is estimated via a first-order analytical model (Supplementary Methods) from operator counts and literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ); these are analytical placeholders for relative comparison, not measured silicon values.

6 Experimental Setup

We evaluate ResNet-18 on CIFAR-10 as an entry validation platform, ConvNeXt-Tiny trained from scratch as the higher-capacity CNN testbed, and Tiny-ViT-5M fine-tuned from ImageNet as the deployment-oriented transformer setting. CIFAR-10, CIFAR-100, and Flowers-102 provide a progression from easier to harder classification settings and from relatively data-rich to data-limited regimes.

The experiments fall into three groups: ResNet baseline quantization and front-end trends, ConvNeXt retention and proportional-noise studies, and Tiny-ViT hardware-aware training together with physical-extension tests such as nonlinear write and fresh-instance transfer. Detailed optimization settings and the Monte Carlo evaluation protocol are summarized in the Supplementary Information.

For ConvNeXt, C1–C4 denote the analogous FP32, quantized/no-noise, noisy-baseline, and hardware-aware training runs; C4 also anchors the proportional-noise and retention follow-up studies described in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work, and the quantitative results below should be interpreted as comparative risk rankings rather than absolute deployment predictions.

7 Conclusion

We presented a behavioral simulation framework for analog compute-in-memory inference that disentangles algorithmic generalization failures from physical precision-retention constraints. By comparing ideal pure-quantization baselines against realistic phase-change memory (PCM) physics, the framework serves as a practical decision aid for establishing deployment frontiers in low-precision analog hardware.

Several observations stand out. At extreme low precision (4-bit), fixed hardware-instance transfer is the most prominent simulated deployment risk observed in our matrix: naive analog deployment collapses to 14.64%, whereas Ensemble Hardware-Aware Training (HAT) resamples

mismatch masks to rescue convergence, recovering fresh-instance accuracy to 86.16±0.19%. Under the tested PCM UnitCell simulation regime, convergence is observed across the 4/6/8-bit precision ladder where pure quantization fails. However, this convergence exposes a precision-retention tradeoff. While 8-bit PCM is drift-flat and 4-bit PCM reaches the most aggressive tested compression at the cost of measurable time-dependent drift, 6-bit emerges as the best observed Pareto midpoint in this sweep, maintaining 8-bit-like stability alongside robust fresh accuracy.

Ultimately, this framework clarifies that standard fixed-mask training is insufficient for extreme low-precision analog deployment. Ensemble HAT resolves this algorithmic bottleneck, allowing system designers to navigate the subsequent physical limits of the analog precision-retention frontier. Extending these hardware-aware training principles to memory-bound inference components, such as analog KV-cache, presents a compelling future trajectory for large-scale deployment.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source data underlying the main-text and Supplementary figures and tables are packaged in the submission-matched source-data archive together with the corresponding manifests, logs, and figure-generation inputs. A reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and `CITATION.cff` file has been staged for Zenodo deposition and will be assigned a DOI at acceptance.

Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at https://github.com/Leslie360/HAT upon acceptance. The current repository is distributed under the Apache-2.0 licence, and the staged Zenodo archive will be updated with the final public DOI at acceptance.

Competing Interests

The authors declare no competing interests.

References

[1] Mark Horowitz. Computing's energy problem (and what we can do about it). In *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323. URL https://doi.org/10.1109/ISSCC.2014.6757323.

[2] Xiaochen Peng, Shanshi Huang, Hongwu Jiang, Anni Lu, and Shimeng Yu. DNN+NeuroSim v2.0: An end-to-end benchmarking framework for compute-in-memory accelerators for on-chip training. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(11):2306–2319, 2021. doi: 10.1109/TCAD.2020.3043731. URL https://doi.org/10.1109/TCAD.2020.3043731.

[3] Mario Lanza et al. The growing memristor industry. *Nature*, 640:613–622, 2025. doi: 10.1038/s41586-025-08733-5.

[4] Malte J. Rasch, Diego Moreda, Tayfun Gokmen, Manuel Le Gallo, Fabio Carta, Cindy Goldberg, Kaoutar El Maghraoui, Abu Sebastian, and Vijay Narayanan. A flexible and fast PyTorch toolkit for simulating training and inference on analog crossbar arrays, 2021. URL https://arxiv.org/abs/2104.02184.

[5] Sandia National Laboratories. Crosssim: Sandia's simulator for analog ai accelerators. Technical report, Sandia National Laboratories, 2024. URL https://doi.org/10.2172/2585829.

[6] Yunchao Xu, Yuan He, Dongyong Shan, Biao Zeng, and Qian-Xi Ni. Emerging artificial synaptic devices based on organic semiconductors: Molecular design, structure and applications. *ACS Applied Materials & Interfaces*, 2025. doi: 10.1021/acsami.4c17455. URL https://doi.org/10.1021/acsami.4c17455.

[7] Haotian Guo, Jing Guo, Yujing Wang, Hezhen Wang, Simin Cheng, Zehao Wang, Qian Miao, and Xiaomin Xu. An organic optoelectronic synapse with multilevel memory enabled by gate modulation. *ACS Applied Materials & Interfaces*, 2024. doi: 10.1021/acsami.3c19624. URL https://doi.org/10.1021/acsami.3c19624.

[8] Li Zhang, Wei Chen, and Jing Xu. Recent progress in organic optoelectronic synaptic devices. *Photonics*, 12(5):435, 2025. doi: 10.3390/photonics12050435. URL https://doi.org/10.3390/photonics12050435.

[9] Corey Lammie, Wei Xiang, Bernabé Linares-Barranco, and Mostafa Rahimi Azghadi. Memtorch: An open-source simulation framework for memristive deep learning systems. *Neurocomputing*, 485:124–133, 2022. doi: 10.1016/j.neucom.2022.02.043. URL https://doi.org/10.1016/j.neucom.2022.02.043.

[10] Jianmin Zeng, Xinhui Chen, Shuzhi Liu, Qilai Chen, and Gang Liu. Organic memristor with synaptic plasticity for neuromorphic computing applications. *Nanomaterials*, 13(5): 803, 2023. doi: 10.3390/nano13050803. URL https://doi.org/10.3390/nano13050803.

[11] Uihoon Jung, Miseong Kim, Jaewon Jang, Jin-Hyuk Bae, In Man Kang, and Sin-Hyung Lee. Formation of cluster-structured metallic filaments in organic memristors for wearable neuromorphic systems with bio-mimetic synaptic weight distributions. *Advanced Science*, 11 (9):2307494, 2024. doi: 10.1002/advs.202307494. URL https://doi.org/10.1002/advs.202307494.

[12] Yu-Pu Lin, Christopher H. Bennett, Théo Cabaret, Damir Vodenicarevic, Djaafar Chabi, Damien Querlioz, Bruno Jousselme, Vincent Derycke, and Jacques-Olivier Klein. Physical realization of a supervised learning system built with organic memristive synapses. *Scientific Reports*, 2016. doi: 10.1038/srep31932. URL https://doi.org/10.1038/srep31932.

[13] Vinay Joshi, Manuel Le Gallo, Simon Haefeli, Irem Boybat, S. R. Nandakumar, Christophe Piveteau, Martino Dazzi, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Accurate deep neural network inference using computational phase-change memory. *Nature Communications*, 11:2473, 2020. doi: 10.1038/s41467-020-16108-9. URL https://doi.org/10.1038/s41467-020-16108-9.

[14] Jungwook Choi, Zhuo Wang, Swagath Venkataramani, Pierce I-Jen Chuang, Vijayalakshmi Srinivasan, and Kailash Gopalakrishnan. PACT: Parameterized clipping activation for quantized neural networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1028–1037, 2018. doi: 10.48550/arXiv.1805.06085. URL http://proceedings.mlr.press/v80/choi18a.html.

[15] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017. doi: 10.1109/IROS.2017.8202133. URL https://doi.org/10.1109/IROS.2017.8202133.

[16] Pietro Belleri, Judith Pons i Tarrès, Iain McCulloch, Paul W. M. Blom, Zsolt M. Kovács-Vajna, Paschalis Gkoupidenis, and Fabrizio Torricelli. Unravelling the operation of organic artificial neurons for neuromorphic bioelectronics. *Nature Communications*, 15:5350, 2024. doi: 10.1038/s41467-024-49668-1. URL https://doi.org/10.1038/s41467-024-49668-1.

[17] Junpeng Ji, Dace Gao, Han-Yan Wu, Miao Xiong, Nevena Stajkovic, Claudia Latte Bovio, Chi-Yuan Yang, Francesca Santoro, Deyu Tu, and Simone Fabiano. Single-transistor organic electrochemical neurons. *Nature Communications*, 16:4334, 2025. doi: 10.1038/s41467-025-59587-4. URL https://doi.org/10.1038/s41467-025-59587-4.

[18] Lan Shen Hu, Marco Fattori, Winston Schilp, Roy Verbeek, Setareh Kazemzadeh, Yoeri van de Burgt, Auke Jisk Kronemeijer, Gerwin Gelinck, and Eugenio Cantatore. An energy-efficient solid-state organic device array for neuromorphic computing. *IEEE Transactions on Electron Devices*, 70(12):6520–6525, 2023. doi: 10.1109/TED.2023.3327947. URL https://doi.org/10.1109/TED.2023.3327947.

[19] Xu Liu, Shilei Dai, Yiyang Jin, Junyao Zhang, Ziyi Guo, Tongrui Sun, Li Li, Pu Guo, Huaiyu Gao, Haixia Liang, Shiqi Zhang, Lize Xiong, Yanmin Zhou, and Jia Huang. Near-infrared organic photoelectrochemical synaptic transistors by wafer-scale photolithography for neuromorphic visual system. *Nature Communications*, 17:197, 2026. doi: 10.1038/s41467-025-66891-6. URL https://doi.org/10.1038/s41467-025-66891-6. Published online 2025-12-08.

[20] Fanqing Zhang, Xinyi Shan, Zhenhua Ji, Yan Wang, Jing Zhao, and Zhongrui Wang. Large-scale high uniform optoelectronic synapses array for artificial visual neural network. *Microsystems & Nanoengineering*, 11:42, 2025. doi: 10.1038/s41378-024-00859-2. URL https://doi.org/10.1038/s41378-024-00859-2.

[21] Heyi Huang, Xiangpeng Liang, Yuyan Wang, Jianshi Tang, Yuankun Li, Yiwei Du, Wen Sun, Jianing Zhang, Peng Yao, Xing Mou, Feng Xu, Jinzhi Zhang, Yuyao Lu, Zhengwu Liu, Jianlin Wang, Zhixing Jiang, Ruofei Hu, Ze Wang, Qingtian Zhang, Bin Gao, Xuedong Bai, Lu Fang, Qionghai Dai, Huaxiang Yin, He Qian, and Huaqiang Wu. Fully integrated multi-mode optoelectronic memristor array for diversified in-sensor computing. *Nature Nanotechnology*, 20(1):93–103, 2025. doi: 10.1038/s41565-024-01794-z. URL https://doi.org/10.1038/s41565-024-01794-z.

[22] Ying Liu, Biao Wang, Lei Wu, Lulu Huang, Lu Lin, Li Xiang, Dongqing Liu, Shiguo Zhang, Chenguang Zhu, and Yijie Tao. Artificial visual synaptic architecture with high-linearity light-modulated weight for optoelectronic neuromorphic computing. *ACS Applied Materials & Interfaces*, 2023. doi: 10.1021/acsami.3c11495. URL https://doi.org/10.1021/acsami.3c11495.

[23] Dingwei Li, Yitong Chen, Huihui Ren, Yingjie Tang, Siyu Zhang, Yan Wang, Lixiang Xing, Qi Huang, Lei Meng, and Bowen Zhu. An active-matrix synaptic phototransistor array for in-sensor spectral processing. *Advanced Science*, 11, 2024. doi: 10.1002/advs.202406401. URL https://doi.org/10.1002/advs.202406401.

[24] Tianqing Wan, Bangjie Shao, Sijie Ma, Yue Zhou, Qiao Li, and Yang Chai. In-sensor computing: Materials, devices, and integration technologies. *Advanced Materials*, 35(37):

2203830, 2023. doi: 10.1002/adma.202203830. URL https://doi.org/10.1002/adma.202203830.

[25] A. Melianas, T. J. Quill, G. LeCroy, Y. Tuchman, H. v. Loo, S. T. Keene, A. Giovannitti, H. R. Lee, I. P. Maria, I. McCulloch, and A. Salleo. Temperature-resilient solid-state organic artificial synapses for neuromorphic computing. *Science Advances*, 6:eabb2958, 2020. doi: 10.1126/sciadv.abb2958. URL https://doi.org/10.1126/sciadv.abb2958.

[26] Ziyi Guo, Junyao Zhang, Ben Yang, Li Li, Xu Liu, Yutong Xu, Yue Wu, Pu Guo, Tongrui Sun, Shilei Dai, Haixia Liang, Jun Wang, Yidong Zou, Lize Xiong, and Jia Huang. Organic high-temperature synaptic phototransistors for energy-efficient neuromorphic computing. *Advanced Materials*, 36:2310155, 2024. doi: 10.1002/adma.202310155. URL https://doi.org/10.1002/adma.202310155.

[27] Xueying Wu, Edward Hanson, Nansu Wang, Qilin Zheng, Xiaoxuan Yang, Huanrui Yang, Shiyu Li, Feng Cheng, Partha Pratim Pande, and Janardhan Rao Doppa. Block-wise mixed-precision quantization: Enabling high efficiency for practical ReRAM-based DNN accelerators. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024. doi: 10.1109/TCAD.2024.3409193. URL https://doi.org/10.1109/TCAD.2024.3409193.

[28] Chenyu Wang, Zhen Dong, Daquan Zhou, Zhenhua Zhu, Yu Wang, Jiashi Feng, and Kurt Keutzer. EPIM: Efficient processing-in-memory accelerators based on epitome. In *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC)*, pages 1–6, 2024. doi: 10.1145/3649329.3657377. URL https://arxiv.org/abs/2311.07620.

[29] Mengke Ge, Junpeng Wang, Binhan Chen, Yingjian Zhong, Haitao Du, Song Chen, and Yi Kang. Allspark: Workload orchestration for visual transformers on processing in-memory systems. *IEEE Transactions on Computers*, 2025. doi: 10.1109/TC.2024.3483633. URL https://doi.org/10.1109/TC.2024.3483633.

[30] Cong Wang, Zexin Fu, Jiayi Huang, and Shanshi Huang. Hemlet: A heterogeneous compute-in-memory chiplet architecture for vision transformers with group-level parallelism, 2025. URL https://arxiv.org/abs/2511.15397.

[31] Shiwei Liu, Chen Mu, Hao Jiang, Yunzhengmao Wang, Jinshan Zhang, Feng Lin, Keji Zhou, Qi Liu, and Chixiao Chen. HARDSEA: Hybrid analog-ReRAM clustering and digital-SRAM in-memory computing accelerator for dynamic sparse self-attention in transformer. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 32(2):269–282, 2024. doi: 10.1109/TVLSI.2023.3337777. URL https://doi.org/10.1109/TVLSI.2023.3337777.

[32] W. Wan, R. Kubendran, B. Gao, S. Joshi, P. Raina, H. Wu, G. Cauwenberghs, and H.-S. P. Wong. A voltage-mode sensing scheme with differential-row weight mapping for energy-efficient RRAM-based in-memory computing. In *2020 IEEE Symposium on VLSI Technology*, pages 1–2, 2020. doi: 10.1109/VLSITechnology18217.2020.9265066. URL https://doi.org/10.1109/VLSITechnology18217.2020.9265066.

[33] Xianggao Wang, Na Wei, Shifan Gao, Wenhao Wu, and Yi Zhao. Accuracy improvement with weight mapping strategy and output transformation for STT-MRAM-based computing-in-memory. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, 10:75–81, 2024. doi: 10.1109/JXCDC.2024.3478360. URL https://doi.org/10.1109/JXCDC.2024.3478360.

[34] Tomas Vincze, Michal Hanic, Martin Berki, and Martin Weis. Simultaneous dual-plasticity organic synaptic transistor for neuromorphic computing. *Advanced Electronic Materials*,

12(1):e00515, 2026. doi: 10.1002/aelm.202500515. URL https://doi.org/10.1002/aelm.202500515. Published online 2025-12-03.